XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 12: The First Law of Thermodynamics

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In this chapter, you'll learn…

- how to calculate work done by a system when its volume changes.
- how to interpret and use the first law of thermodynamics.
- four important kinds of thermodynamic processes.
- why the internal energy of an ideal gas depends on temperature only.
- the difference between molar heat capacities at constant volume and at constant pressure.
- how to analyze adiabatic processes in an ideal gas.

Thermodynamics 热力学

- Thermodynamics is the study of the transformation between heat and other forms of energy.
- We will revisit the conservation of energy in the form of the first law of thermodynamics.
- Examples of thermodynamics:



火车头





- A steam locomotive operates using the laws of thermodynamics, but so do air conditioners and rockets. 类比推理

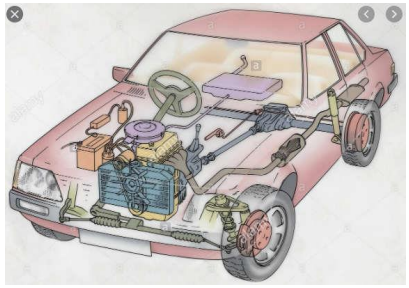
Thermodynamic System

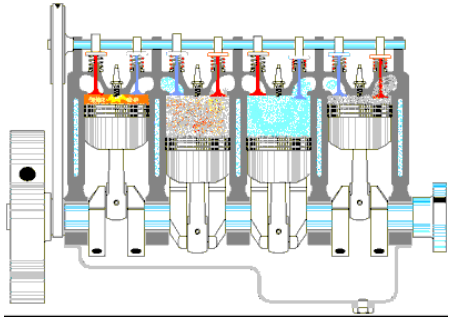
- A thermodynamic system is any collection of objects that may exchange energy with its surroundings.
- When there are changes in the state of a thermodynamic system, is called a thermodynamic process.



- The popcorn in the pot is a thermodynamic system. In the thermodynamic process shown here, heat is added to the popcorn/system, and the system does work on its system surroundings to lift the lid of the pot.

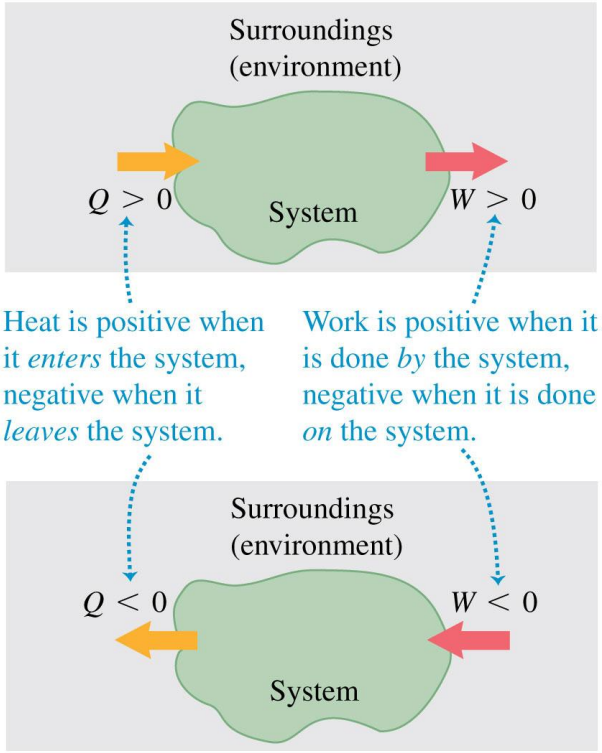
- In a car engine heat is generated by the chemical reaction of oxygen and vaporized gasoline in the engine's cylinders.
- The heated gas pushes on the pistons within the cylinders, doing mechanical work that is used to propel the car.





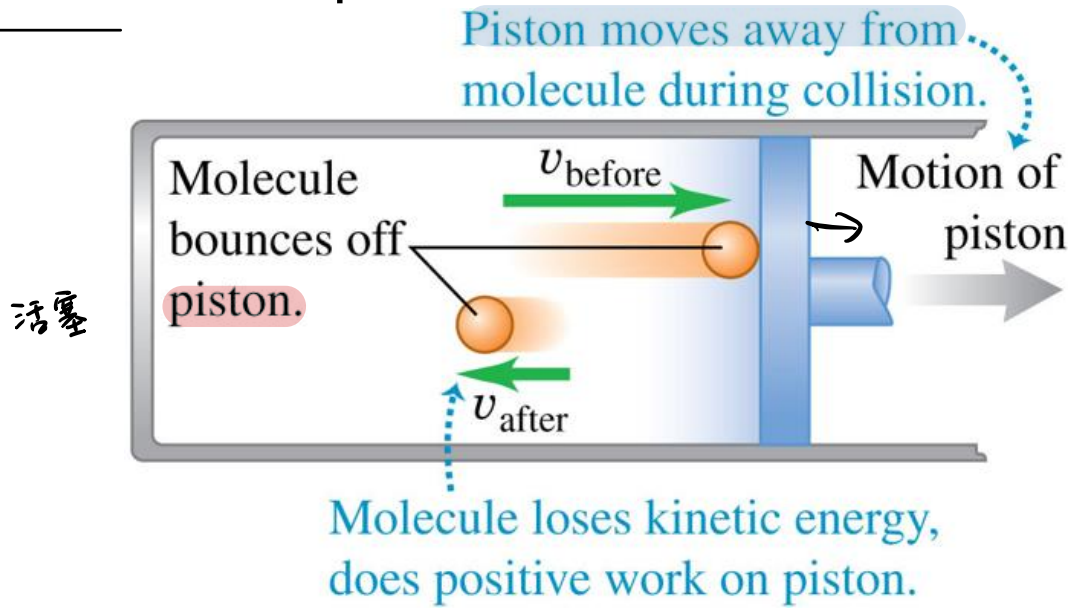
Thermodynamic Systems

- In a thermodynamic process, changes occur in the state of the system. Sign matters!
  - A positive heat, Q represents heat flowing into the system, with a corresponding input of energy to it;
  - A negative Q represents heat flowing out of the system.
  - A positive work, W is the work done by the system, such as work done by an expanding gas, represents energy leaving the system;
  - A negative W indicates work done on the system by its surroundings, such as work done during compression of a gas by its surroundings, represents energy entering the system.



Work Done During Volume Changes

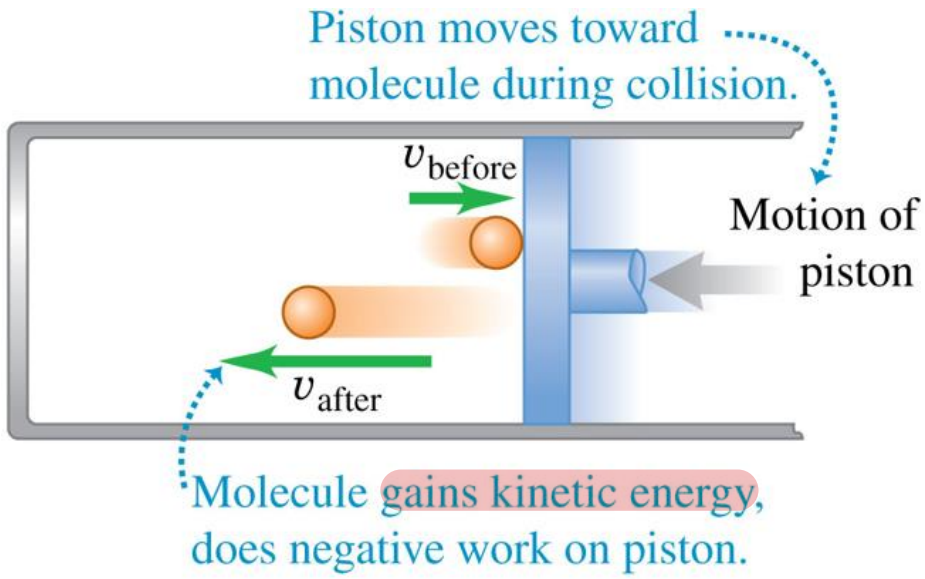
- We can understand the work done by a gas in a volume change by considering a molecule in the gas.
- When one such molecule collides with a surface moving to the right, so the volume of the gas increases, the molecule does positive work on the piston.



活塞

Work Done During Volume Changes

- If the piston moves toward the left as in the figure shown here, so the volume of the gas decreases; positive work is done on the molecule during the collision.
- Hence the gas molecules do negative work on the piston.



气体对外做功

$V>0$, $W>0$.

Work Done During Volume Changes

- The infinitesimal work done by the system during the small expansion dx is $dW = pA\,dx$.
- In a finite change of volume from V_1 to V_2:

$p = \frac{F}{A}$ $F = pA$

$dW = F dx$

(19.2) $= pA\,dx$

Work done in a volume change $W = \int_{V_1}^{V_2} p\,dV$

Upper limit = final volume

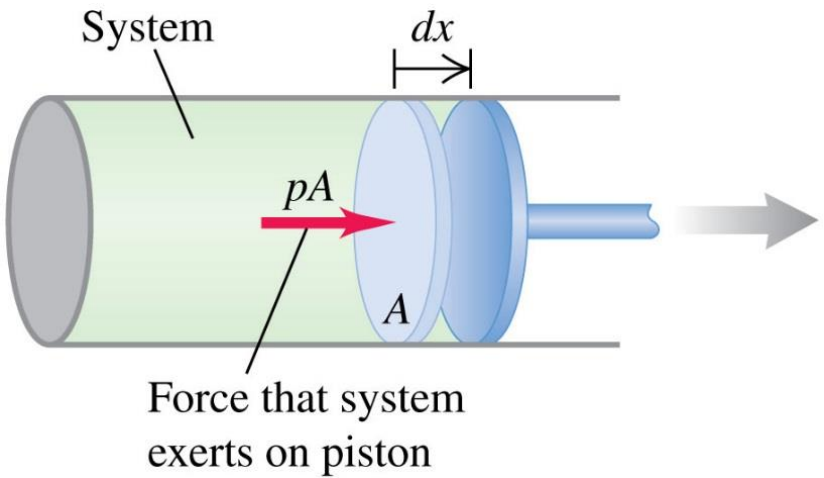
Integral of the pressure with respect to volume

Lower limit = initial volume

引入微分求W.

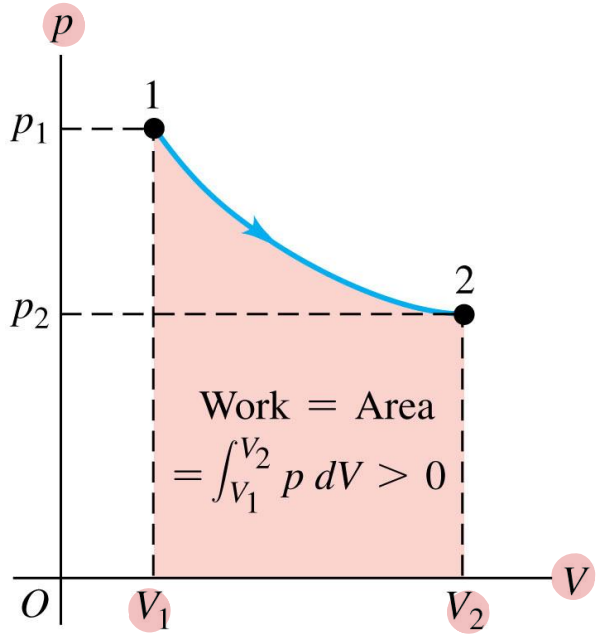
$W = \int_{V_1}^{V_2} p\,dV$

由 $F \cdot x = W$ 导的



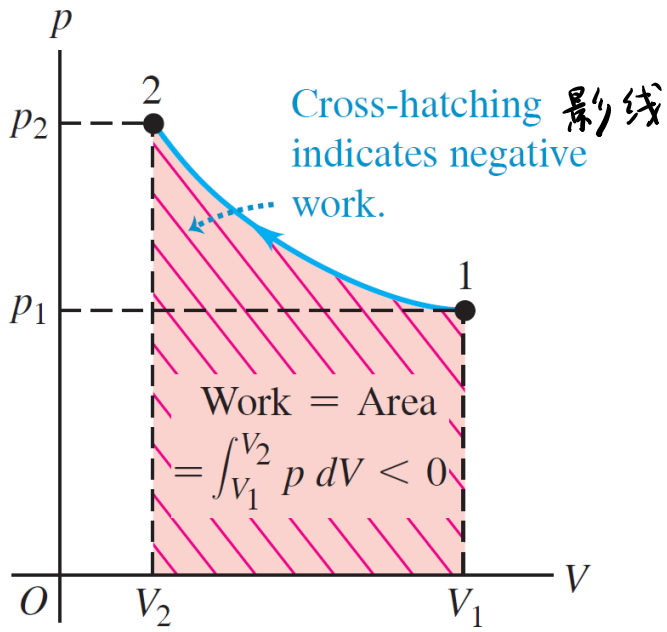
Work Done During Volume Changes

- The work done equals the area under the curve on a pV-diagram.
- Shown in the graph is a system undergoing an expansion with varying pressure.



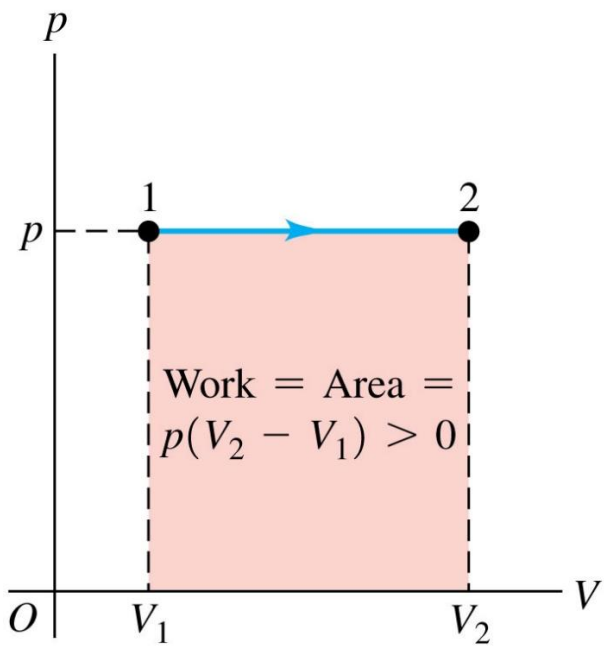
Work Done During Volume Changes

- Shown in the graph is a system undergoing a compression with varying pressure. 压缩
- In this case the work is negative.



Work Done During Volume Changes

- Shown in the graph is a system undergoing an expansion with constant pressure.
- In this case, $W = p(V_2 - V_1)$ (19.3)



气体对外做 正 负 0 功的图像

Work Done During Volume Changes

EXAMPLE 19.1 ISOTHERMAL EXPANSION OF AN IDEAL GAS

As an ideal gas undergoes an *isothermal* (constant-temperature) expansion at temperature T, its volume changes from V_1 to V_2. How much work does the gas do?

Constant-temperature T.

$p = \frac{nRT}{V}$

$\int_{V_1}^{V_2} \frac{dV}{V} = \ln\frac{V_2}{V_1}$

$W = \int_{V_1}^{V_2} p\,dV = \int_{V_1}^{V_2} \frac{nRT}{V}\,dV = nRT\left(-\frac{1}{2V^2}\right)\Big|_{V_1}^{V_2} = nRT\left(-\frac{1}{2V_2^2} + \frac{1}{2V_1^2}\right)$

EXECUTE: From Eq. (18.3),

$$p = \frac{nRT}{V}$$

We substitute this into the integral of Eq. (19.2), take the constant factor nRT outside, and evaluate the integral:

$$W = \int_{V_1}^{V_2} p\,dV = nRT\int_{V_1}^{V_2}\frac{dV}{V}$$

$$= nRT\ln\frac{V_2}{V_1} \quad \text{(ideal gas, isothermal process)}$$

We can rewrite this expression for W in terms of p_1 and p_2. Because $pV = nRT$ is constant,

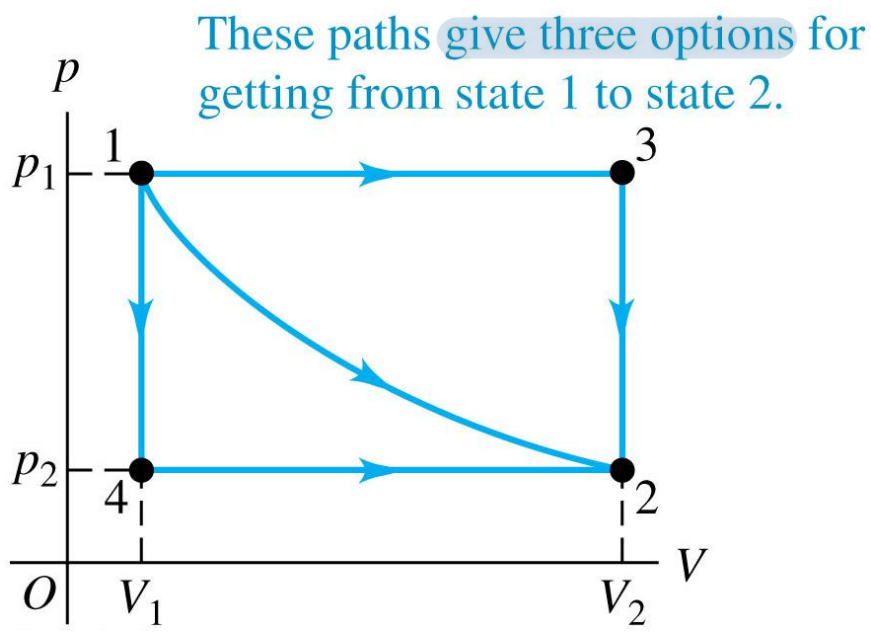
$$p_1V_1 = p_2V_2 \qquad \text{or} \qquad \frac{V_2}{V_1} = \frac{p_1}{p_2}$$

so

$$W = nRT\ln\frac{p_1}{p_2} \quad \text{(ideal gas, isothermal process)}$$

Paths Between Thermodynamic States

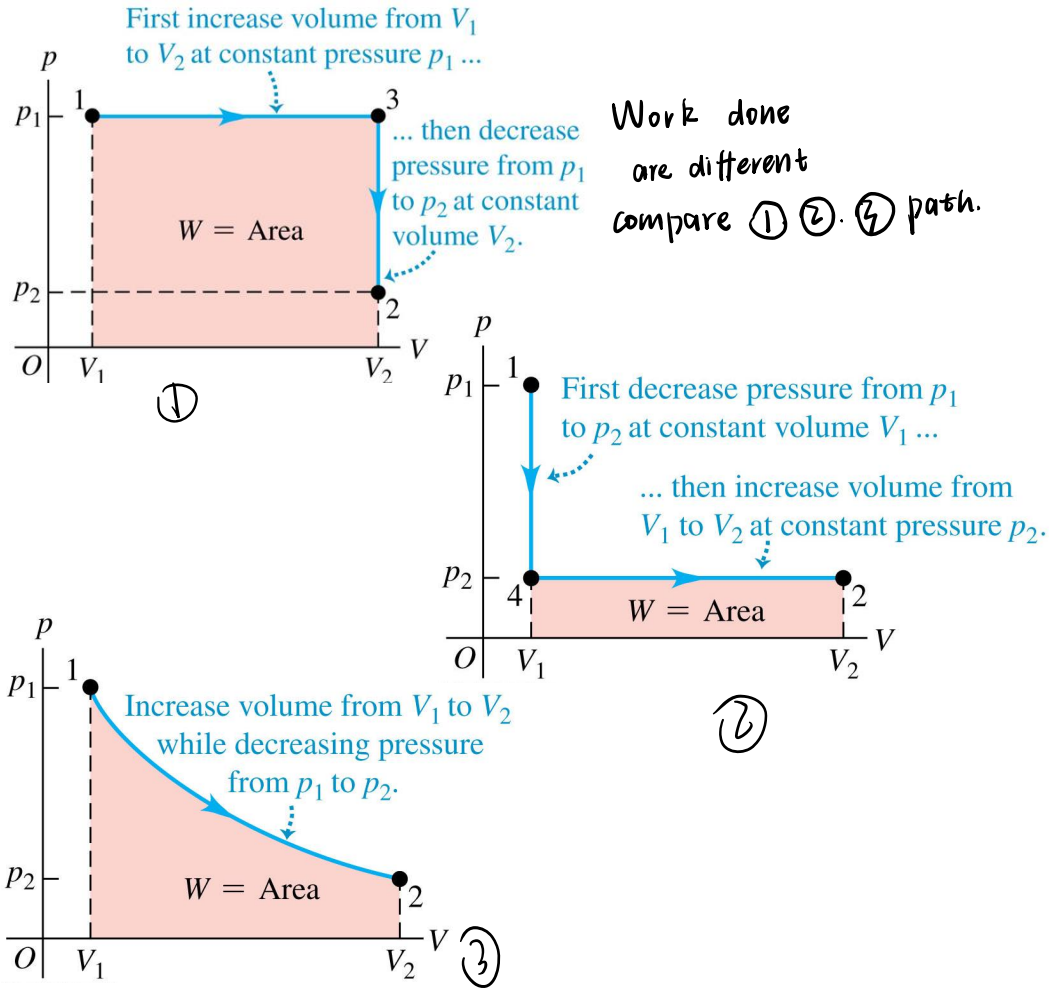
- When a thermodynamic system changes from an initial state to a final state, it passes through a series of intermediate states. We call this series of states a path.
- The work done by a system during a transition between two states depends on the path chosen.
- Consider three different paths on a pV-diagram for getting from state 1 to state 2.



不同 path，做功 different

Paths Between Thermodynamic States

- If $1 \rightarrow 3 \rightarrow 2$. The system does a large amount of work under the path.
- If $1 \rightarrow 4 \rightarrow 2$. The system does a small amount of work under the path
- If along the smooth curve from 1 to 2, the work done is different from that for either of the other paths.



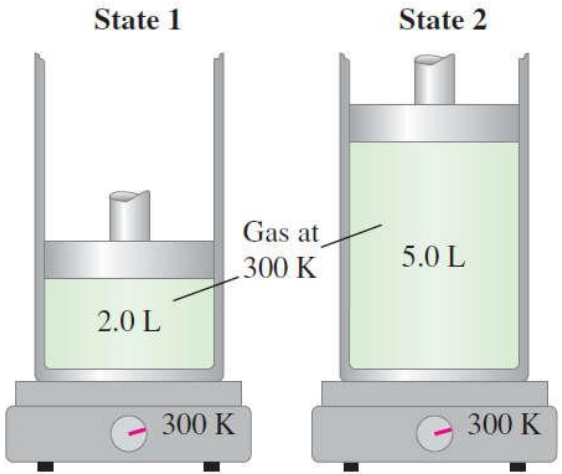
Heat Added in A Thermodynamic Process

- Like work, the heat added to a thermodynamic system when it undergoes a change of state depends on the path from the initial state to the final state. The two different ways for doing experiments.

1. First Experiment

— To change the volume of a certain quantity of an ideal gas from 2.0 L to 5.0 L while keeping the temperature constant at T = 300K.

— let the gas expand slowly, supplying heat from the electric heater to keep the temperature at 300 K. After expanding in this slow, controlled, isothermal manner, the gas reaches its final volume of 5.0 L; it absorbs a definite amount of heat in the process.



System does work on piston; hot plate adds heat to system, $W > 0$ and $Q > 0$.

Heat Added in A Thermodynamic Process

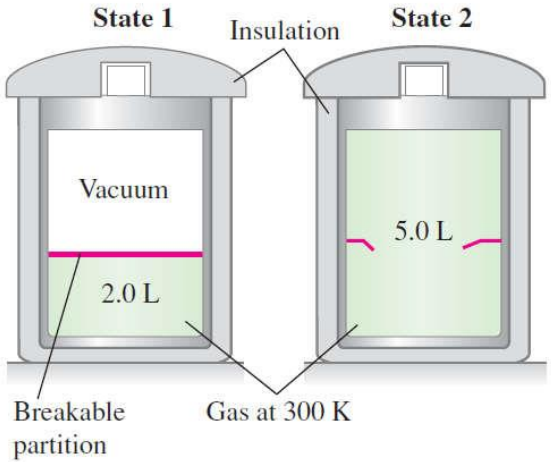
2. Second Experiment - Free expansion experiment

— A different process leading to the same final state. The container is surrounded by insulating walls and is divided by a thin, breakable partition into two compartments.

— Breaking the partition, the gas undergoes a rapid, uncontrolled expansion, with no heat passing through the insulating walls.

— The gas does no work during this expansion because it doesn't push against anything that moves, W = 0. This uncontrolled expansion of a gas into vacuum is called a free expansion.

— The temperature doesn't change, and no heat enters or leaves system, Q = 0.



- The conclusion from these two experiments is that like work, heat depends not only on the initial and final states but also on the path.

功和热 depends on 状态变化和变化路线

First Law of Thermodynamics

- The internal energy, U, of a system is the sum of the kinetic energies of all of its constituent particles, plus the sum of all the potential energies of interaction among these particles.
- The change in the internal energy ΔU of a system is equal to the heat added minus the work done by the system:

First law 就是这个公式

First law of thermodynamics:

$$\Delta U = Q - W \quad (19.4)$$

Internal energy change of thermodynamic system

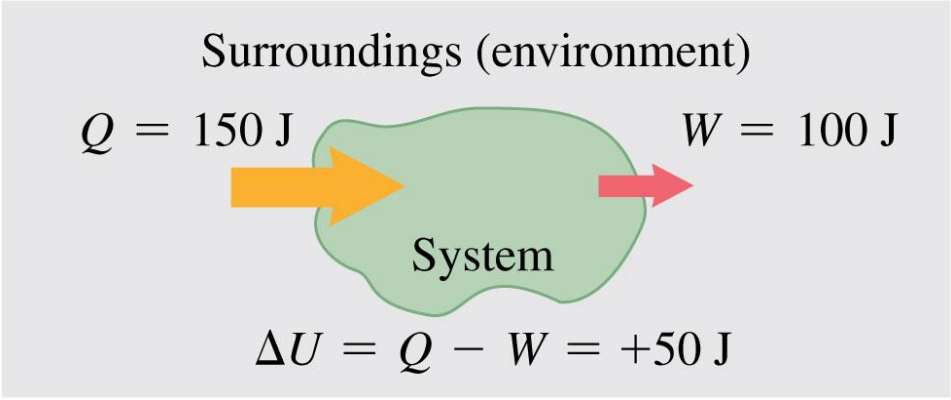
Heat added to system

Work done by system

- The first law of thermodynamics is just a generalization of the conservation of energy. 能量守恒在热力方面的应用 概论
- Both Q and W depend on the path chosen between states, but is independent of the path.
- If the changes are infinitesimal, we write the first law as $dU = dQ - dW$. (19.6)

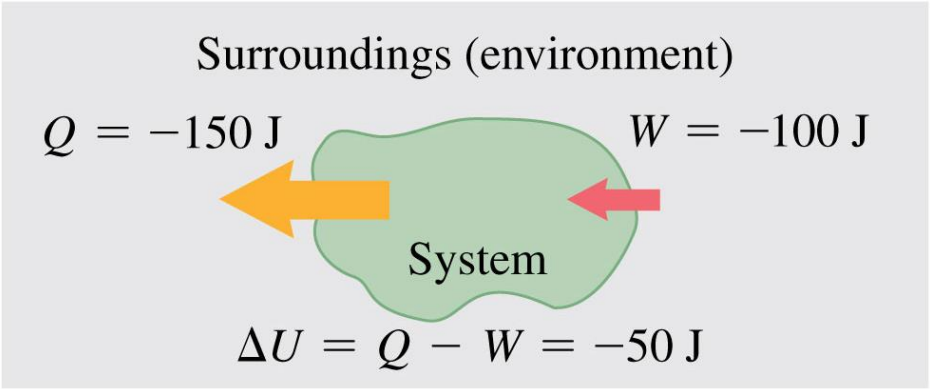
First Law of Thermodynamics

- In a thermodynamic process, the internal energy U of a system may **increase**. ↑
- In the system shown below, more heat is added to the system than the system does work.
- So the internal energy of the system increases.



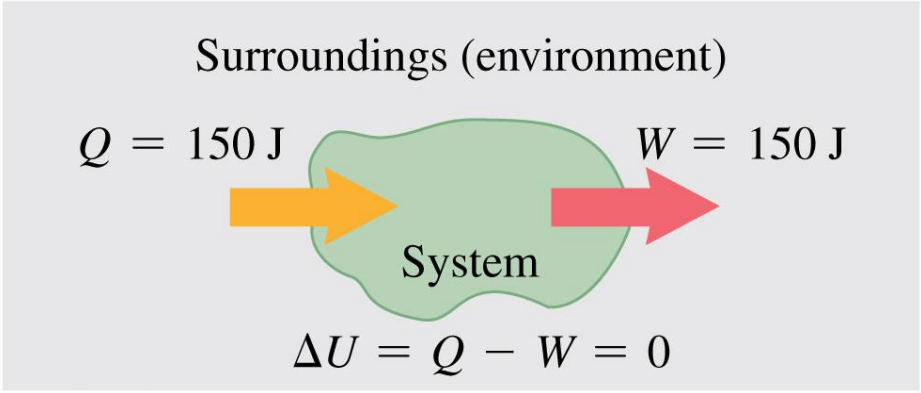
First Law of Thermodynamics

- In a thermodynamic process, the internal energy U of a system may **decrease**. ↓
- In the system shown below, more heat flows out of the system than work is done.
- So the internal energy of the system decreases.



First Law of Thermodynamics

- In a thermodynamic process, the internal energy U of a system may **remain the same**.
- In the system shown below, the heat added to the system equals the work done by the system.
- So the internal energy of the system is unchanged.



解释了内能变化的三种情况

$Q-W>0$

$=0$

<0

First Law of Thermodynamics – Human Body

- Your body is a thermodynamic system.
- When you do a push-up, your body does work, so $W > 0$.
- Your body also warms up during exercise; by perspiration (sweating) and other means the body rids itself of this heat, so $Q < 0$.
- Since Q is negative and W is positive, $U = Q - W < 0$ and the body's internal energy decreases.
- That's why exercise helps you lose weight: It uses up some of the internal energy stored in your body in the form of fat.



実例

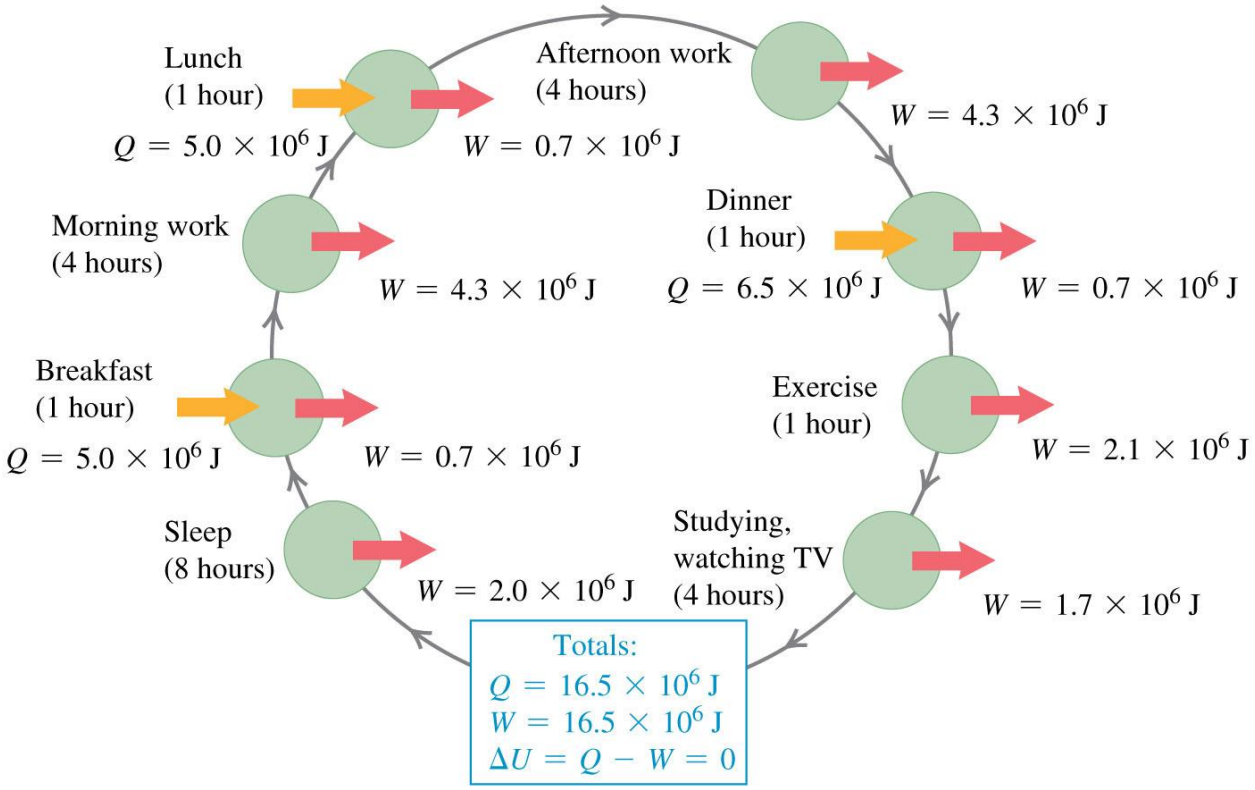
First Law of Thermodynamics – Human Body

- The internal energy of a cup of coffee depends on just its thermodynamic state—how much water and ground coffee it contains, and what its temperature is.
- It does not depend on the history of how the coffee was prepared—that is, the thermodynamic path that led to its current state.



First Law of Thermodynamics – Human Body

- Cyclic Thermodynamic Process for a Human Day



First Law of Thermodynamics

$\Delta U = 0 = Q - W$

$W = mgh = 60 \times 9.8 \times h$

$\|$

$Q = 900\ \text{kcal} \times 4186\ \text{J}/1\ \text{kcal}$

$= 3.77 \times 10^6\ \text{J}$

EXAMPLE 19.2 WORKING OFF YOUR DESSERT

You propose to climb several flights of stairs to work off the energy you took in by eating a 900-calorie hot fudge sundae. How high must you climb? Assume that your mass is 60.0 kg.

EXECUTE: From the first law of thermodynamics, $\Delta U = 0 = Q - W$, so $W = mgh = Q$. Hence you must climb to height $h = Q/mg$. First convert units: $Q = (900\ \text{kcal})(4186\ \text{J}/1\ \text{kcal}) = 3.77 \times 10^6\ \text{J}$. Then

$$h = \frac{Q}{mg} = \frac{3.77 \times 10^6\ \text{J}}{(60.0\ \text{kg})(9.80\ \text{m/s}^2)} = 6410\ \text{m}$$

EVALUATE: We have unrealistically assumed 100% efficiency in the conversion of food energy into mechanical work. The actual efficiency is roughly 25%, so the work W you do as you "burn off" the sundae is only about $(0.25)(900\ \text{kcal}) = 225\ \text{kcal}$. (The remaining 75%, or 675 kcal, is transferred to your surroundings as heat.) Hence you actually must climb about $(0.25)(6410\ \text{m}) = 1600\ \text{m}$, or one *mile*! Do you really want that sundae?

First Law of Thermodynamics

EXAMPLE 19.5 THERMODYNAMICS OF BOILING WATER

One gram of water ($1\ \text{cm}^3$) becomes $1671\ \text{cm}^3$ of steam when boiled at a constant pressure of 1 atm ($1.013 \times 10^5\ \text{Pa}$). The heat of vaporization at this pressure is $L_v = 2.256 \times 10^6\ \text{J/kg}$. Compute (a) the work done by the water when it vaporizes and (b) its increase in internal energy.

$W = p \cdot \Delta V$
$= 1.013 \times 10^5 \times (1671 \times 10^{-6}\ \text{m}^3 - 1 \times 10^{-6}\ \text{m}^3)$
$= 169\ \text{J}$

EXECUTE: (a) From Eq. (19.3), the water does work

$$W = p(V_2 - V_1)$$
$$= (1.013 \times 10^5\ \text{Pa})(1671 \times 10^{-6}\ \text{m}^3 - 1 \times 10^{-6}\ \text{m}^3)$$
$$= 169\ \text{J}$$

(b) From Eq. (17.20), the heat added to the water is

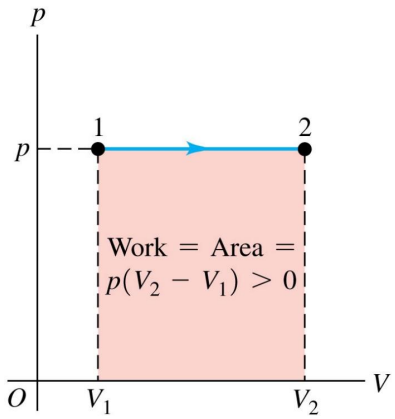
$$Q = mL_v = (10^{-3}\ \text{kg})(2.256 \times 10^6\ \text{J/kg}) = 2256\ \text{J}$$

Then from Eq. (19.4),

$$\Delta U = Q - W = 2256\ \text{J} - 169\ \text{J} = 2087\ \text{J}$$

$Q = mL_v$
$\Delta U = Q - W = mL_v - 169\ \text{J}$

Water density = 997 kg/m³ ≈ 1000 kg/m³



知道，水的密度
1000 kg/m³

Four Kinds of Thermodynamic Processes

- There are four specific kinds of thermodynamic processes that occur often in practical situations:

隔熱 ➢ Adiabatic: No heat is transferred into or out of the system, so $Q = 0$. Also, $\Delta U = U_2 - U_1 = -W$.

等容 ➢ Isochoric: The volume remains constant, so $W = 0$.

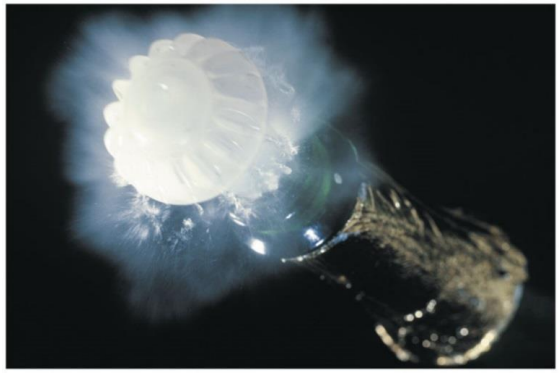
等压 ➢ Isobaric: The pressure remains constant, so $W = p(V_2 - V_1)$.

等温 ➢ Isothermal: The temperature remains constant.

Four Kinds of Thermodynamic Processes

Adiabatic Process 隔熱

- When the cork is popped on a bottle of champagne, the pressurized gases inside the bottle expand rapidly and do positive work on the outside air.
- There is little time for the gases to exchange heat with their surroundings, so the expansion is nearly adiabatic.
- Hence the internal energy of the expanding gases decreases ($\Delta U = -W$) and their temperature drops.
- This makes water vapor condense and form a miniature cloud.



Four Kinds of Thermodynamic Processes

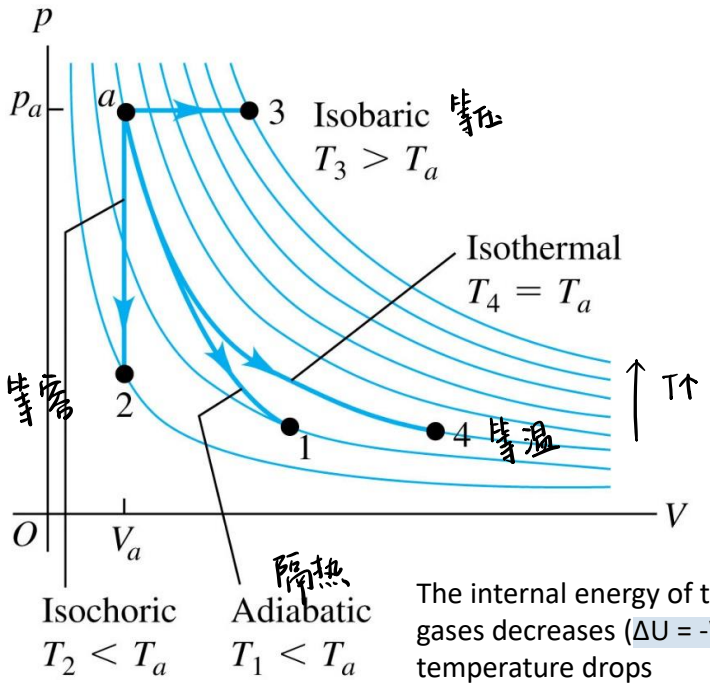
Isobaric Process

- Most cooking involves isobaric processes.
- That's because the air pressure above a saucepan or frying pan, or inside a microwave oven, remains essentially constant while the food is being heated.



Four Kinds of Thermodynamic Processes

- Shown are the paths on a pV-diagram for all four different processes for a constant amount of an ideal gas, all starting at state a.



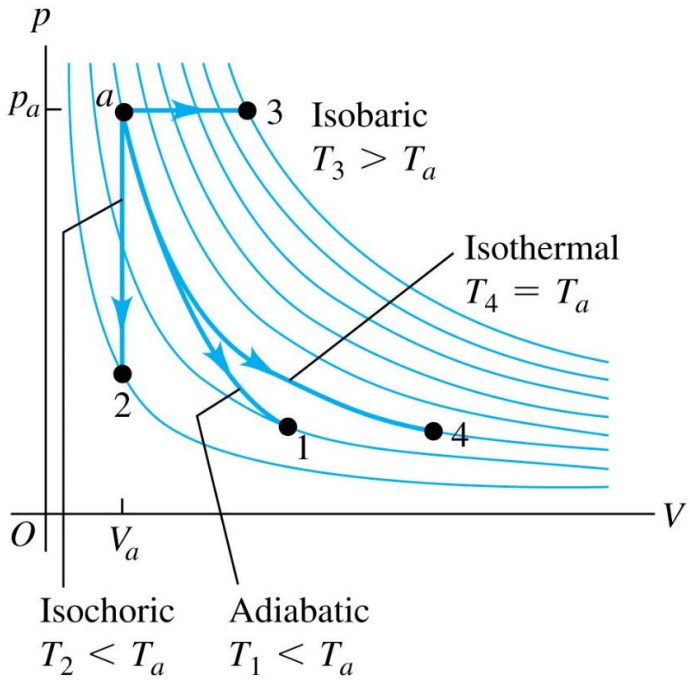


The internal energy of the expanding gases decreases ($\Delta U = -W$) and their temperature drops

Four Kinds of Thermodynamic Processes

- Shown are the paths on a pV-diagram for all four different processes for a constant amount of an ideal gas, all starting at state a.



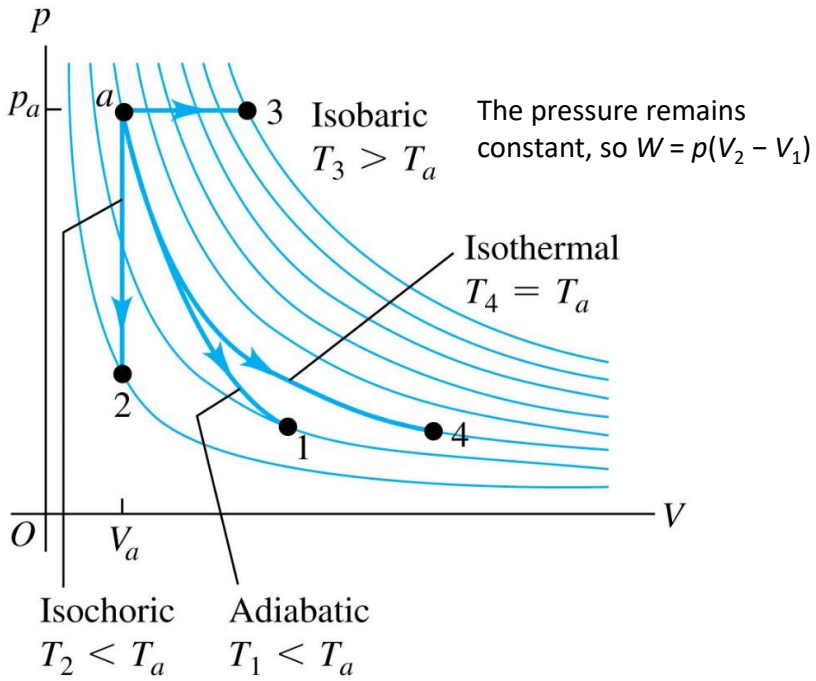


The volume remains constant, so $W = 0$

An example of a nearly isochoric process is a pressure cooker. When sealed closed, the volume inside cannot change, so when heat is added both pressure and temperature increase rapidly. In actuality, pressure cookers do expand slightly, and some gas is released from a valve on top.

Four Kinds of Thermodynamic Processes

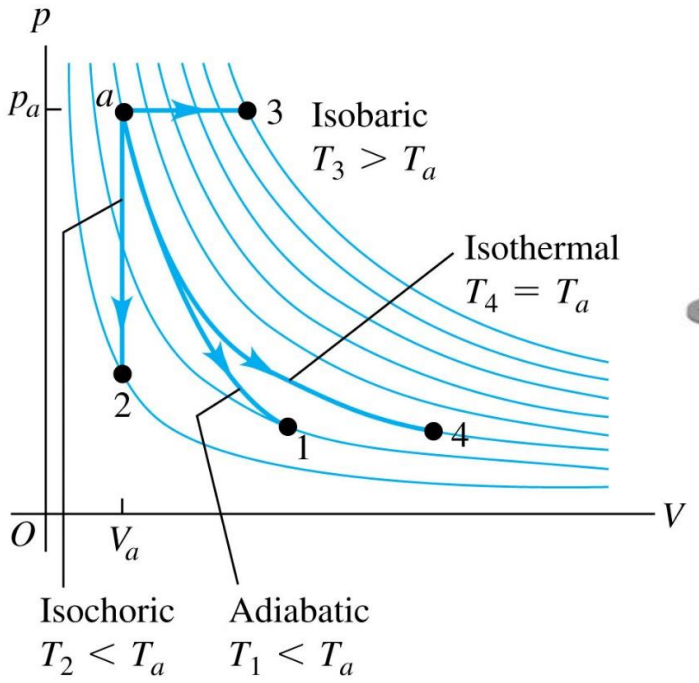
- Shown are the paths on a pV-diagram for all four different processes for a constant amount of an ideal gas, all starting at state a.

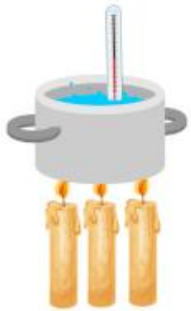




Four Kinds of Thermodynamic Processes

- Shown are the paths on a pV-diagram for all four different processes for a constant amount of an ideal gas, all starting at state a.





The process in which the temperature remains constant is known as *Isothermal process.*

理想气体内能只与T有关.

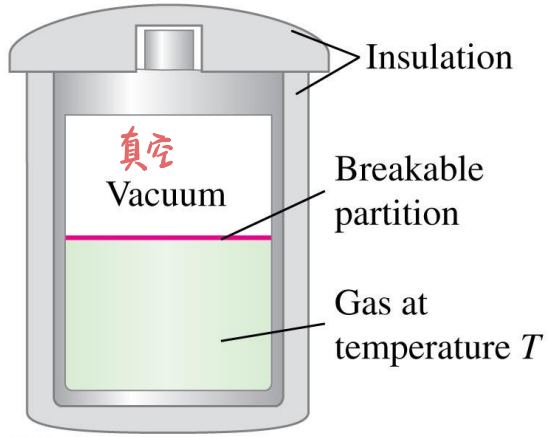
Internal Energy of an Ideal Gas

- The internal energy of an ideal gas depends only on its temperature, not on its pressure or volume.
- The temperature of an ideal gas does not change during a free expansion.

Free expansion experiment

$\frac{PV}{T}$

i. Even when the *p* and *v* change, the internal energy is still constant. $\Delta U = 0 = \underset{=0}{Q} - \underset{=0}{W}$
ii. Temperature is constant in this free expansion experiment.
iii. Many experiments with low-density gases have shown that when a low-density gas undergoes a free expansion, its temperature *does not* change.

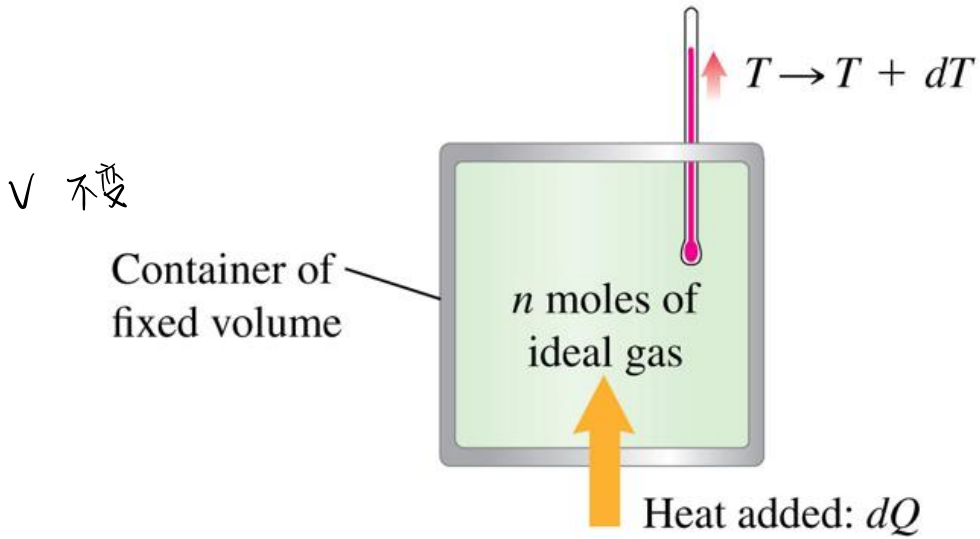


Heat Capacities of an Ideal Gas

- C_V is the molar heat capacity at constant volume.
- To measure C_V, we raise the temperature of an ideal gas in a rigid container with constant volume, ignoring its thermal expansion.

如何测量气体热容量

$$\text{Constant volume: } dQ = nC_V\,dT \quad (19.12)$$



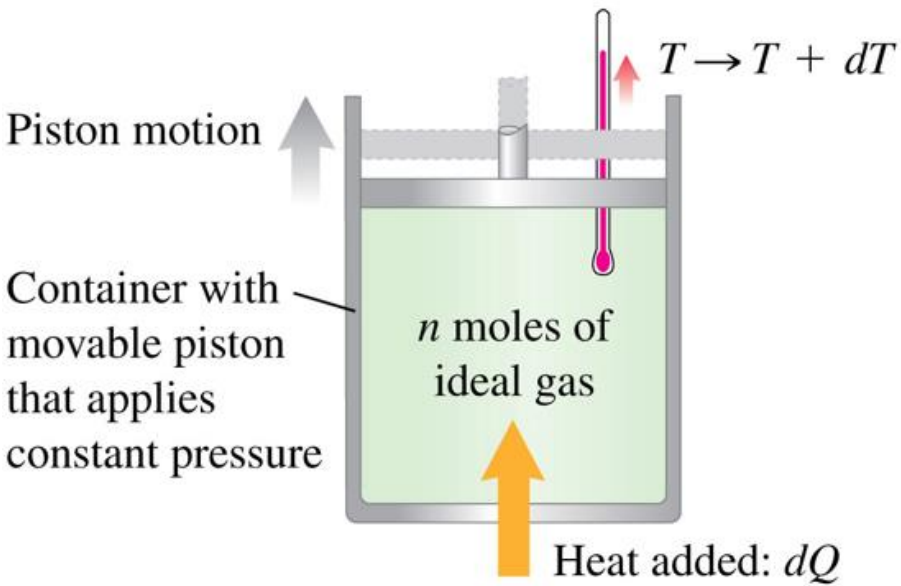
Heat Capacities of an Ideal Gas

- C_p is the molar heat capacity at constant pressure.
- To measure C_p, we let the gas expand just enough to keep the pressure constant as the temperature rises.

理想气体热容量

P-定
V-定

Constant pressure: $dQ = nC_p\,dT$



Relating C_V and C_p for an Ideal Gas

- To produce the same temperature change, more heat is required at constant pressure than at constant volume since ΔU is the same in both cases.
- This means that $C_p > C_V$.

For an ideal gas:

$$C_p = C_V + R \qquad (19.17)$$

(C_p: Molar heat capacity at constant *pressure*; C_V: Molar heat capacity at constant *volume*; R: Gas constant)

- R is the gas constant

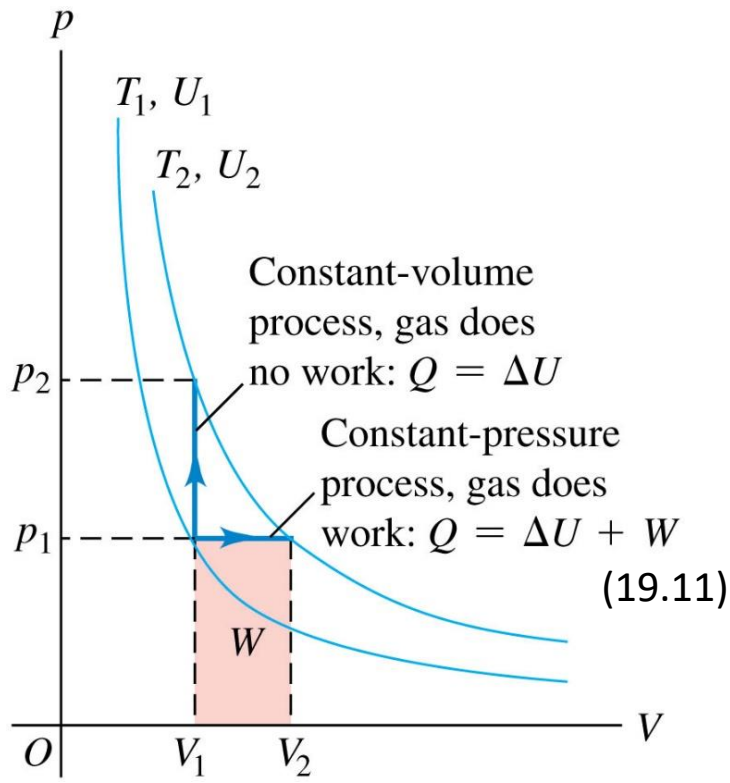
$$R = 8.314 \frac{\text{J}}{\text{mol}} \cdot K.$$

引入气体常量

$dW = pdV = nRdT$

$nC_pdT = dn + nRdT$

All ideal gas



Relating C_V and C_p for an Ideal Gas

- The ratio of heat capacities is:

Ratio of heat capacities $\gamma = \dfrac{C_p}{C_V}$ — C_p: Molar heat capacity at constant *pressure*; C_V: Molar heat capacity at constant *volume*

(19.18)

- For monatomic (单原子) ideal gases, $C_V = \frac{3}{2}R$

$$C_p = C_V + R = \tfrac{3}{2}R + R = \tfrac{5}{2}R \quad \text{so,} \quad \gamma = \frac{C_p}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \tfrac{5}{3} = 1.67$$

- For diatomic (双原子) ideal gases, $C_V = \frac{5}{2}R$

$$C_p = C_V + R = \tfrac{7}{2}R \quad \text{so,} \quad \gamma = \frac{C_p}{C_V} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = \tfrac{7}{5} = 1.40$$

$C_p - C_V = R$

Relating C_V and C_p for an Ideal Gas

EXAMPLE 19.6 COOLING YOUR ROOM

A typical dorm room or bedroom contains about 2500 moles of air. Find the change in the internal energy of this much air when it is cooled from 35.0°C to 26.0°C at a constant pressure of 1.00 atm. Treat the air as an ideal gas with $\gamma = 1.400$.

$$\gamma = \frac{C_p}{C_v} = 1.4 = \frac{C_v + R}{C_v} = 1 + \frac{R}{C_v}$$

$$C_v = \frac{R}{\gamma - 1} = 20.79$$

EXECUTE: From Eqs. (19.17) and (19.18),

$$\gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V}$$

$$C_V = \frac{R}{\gamma - 1} = \frac{8.314\ \text{J/mol} \cdot \text{K}}{1.400 - 1} = 20.79\ \text{J/mol} \cdot \text{K}$$

Then from Eq. (19.13),

$$\Delta U = nC_V\,\Delta T$$

$$= (2500\ \text{mol})(20.79\ \text{J/mol} \cdot \text{K})(26.0°\text{C} - 35.0°\text{C})$$

$$= -4.68 \times 10^5\ \text{J}$$

EVALUATE: To cool 2500 moles of air from 35.0°C to 26.0°C, a room air conditioner must extract this much internal energy from the air and transfer it to the air outside. In Chapter 20 we'll discuss how this is done.

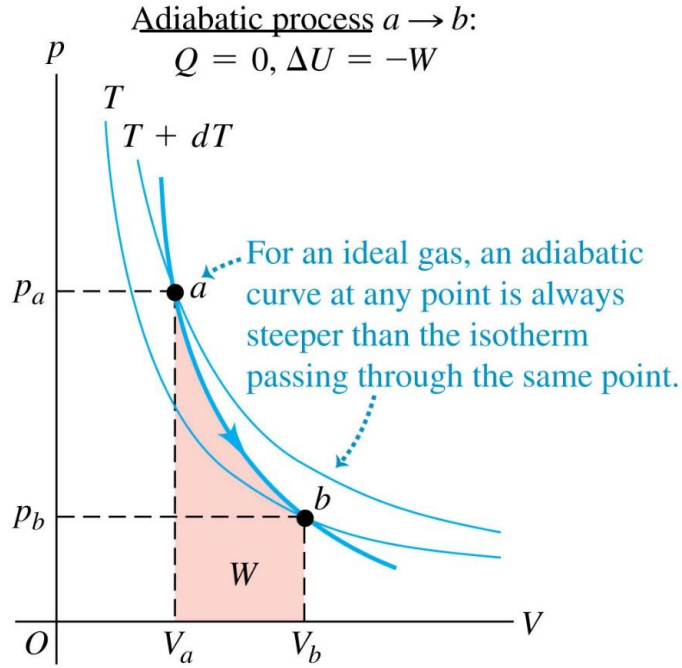
Why C_V not C_p?

One of the special properties of an ideal gas is that its internal energy depends only on temperature. Thus the change in internal energy during any process must be determined only by the temperature change.

隔热 Adiabatic Processes for an Ideal Gas

- In an adiabatic process, no heat is transferred in or out of the gas, so $Q = 0$.
- Shown is a pV–diagram for an adiabatic expansion.
- As the gas expands, it does positive work W on its environment, so its internal energy decreases, and its temperature drops.
- Note that an adiabatic curve at any point is always steeper than an isotherm at that point.

比等温线低，因为会降温



Adiabatic Processes for an Ideal Gas

- We can also calculate the work done by an ideal gas during an adiabatic process.
- We know that Q = 0 and W = -ΔU for any adiabatic process.
- For an ideal gas, $\Delta U = nC_V (T_2 - T_1)$. If the number of moles n and the initial and final temperatures T_1 and T_2 are known, we have simply

Work done by an ideal gas, adiabatic process

$$W = nC_V(T_1 - T_2)$$

Number of moles; Initial temperature; Final temperature; Molar heat capacity at constant volume

(19.25)

- We may also use $pV = nRT$ in this equation to obtain

Work done by an ideal gas, adiabatic process

$$W = \frac{C_V}{R}(p_1V_1 - p_2V_2) = \frac{1}{\gamma - 1}(p_1V_1 - p_2V_2)$$

Molar heat capacity at constant volume; Initial pressure, volume; Gas constant; Ratio of heat capacities; Final pressure, volume

$$C_V = \frac{R}{\gamma - 1}$$

(19.26)

在隔热状态下的功，两个公式.

Adiabatic Processes for an Ideal Gas

- For an initial state (T_1, V_1) and a final state (T_2, V_2):

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \qquad \text{(adiabatic process, ideal gas)} \qquad (19.22)$$

- For an initial state (p_1, V_1) and a final state (p_2, V_2):

$$p_1 V_1^{\gamma} = p_2 V_2^{\gamma} \qquad \text{(adiabatic process, ideal gas)} \qquad (19.24)$$

Adiabatic Processes for an Ideal Gas

- Put your hand a few centimeters in front of your mouth, open your mouth wide, and exhale.
- Your breath will feel warm on your hand, because the exhaled gases emerge 浮现 at roughly the temperature of your body's interior.
- Now purse your lips as though you were going to whistle, and again blow on your hand.
- The exhaled gases will feel much cooler.
- In this case the gases undergo a rapid, essentially adiabatic expansion as they emerge from between your lips, so the temperature of the exhaled gases decreases.



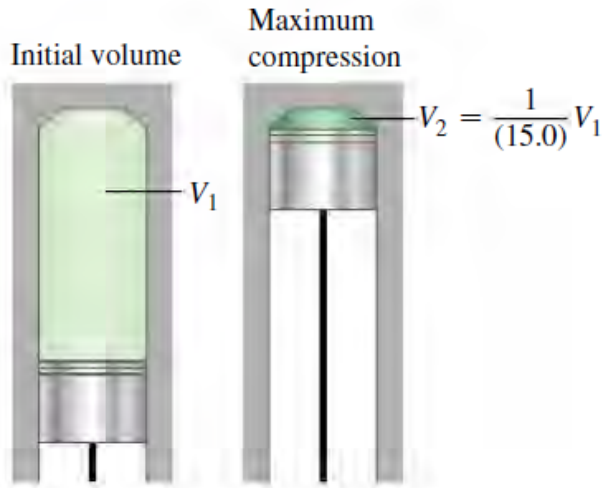


Adiabatic Processes for an Ideal Gas

EXAMPLE 19.7 ADIABATIC COMPRESSION IN A DIESEL ENGINE

The compression ratio of a diesel engine is 15.0 to 1; that is, air in a cylinder is compressed to $\frac{1}{(15.0)}$ of its initial volume (**Fig. 19.21**). (a) If the initial pressure is 1.01×10^5 Pa and the initial temperature is 27°C (300 K), find the final pressure and the temperature after adiabatic compression. (b) How much work does the gas do during the compression if the initial volume of the cylinder is $1.00\ \text{L} = 1.00 \times 10^{-3}\ \text{m}^3$? Use the values $C_V = 20.8\ \text{J/mol} \cdot \text{K}$ and $\gamma = 1.400$ for air.

19.21 Adiabatic compression of air in a cylinder of a diesel engine.



$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = (300\ \text{K})(15.0)^{0.40} = 886\ \text{K} = 613°\text{C}$$

$$p_2 = p_1\left(\frac{V_1}{V_2}\right)^{\gamma} = (1.01 \times 10^5\ \text{Pa})(15.0)^{1.40}$$

$$= 44.8 \times 10^5\ \text{Pa} = 44\ \text{atm}$$

Adiabatic Processes for an Ideal Gas

EXAMPLE 19.7 ADIABATIC COMPRESSION IN A DIESEL ENGINE

The compression ratio of a diesel engine is 15.0 to 1; that is, air in a cylinder is compressed to $\frac{1}{(15.0)}$ of its initial volume (**Fig. 19.21**). (a) If the initial pressure is 1.01×10^5 Pa and the initial temperature is 27°C (300 K), find the final pressure and the temperature after adiabatic compression. (b) How much work does the gas do during the compression if the initial volume of the cylinder is $1.00\ \text{L} = 1.00 \times 10^{-3}\ \text{m}^3$? Use the values $C_V = 20.8\ \text{J/mol} \cdot \text{K}$ and $\gamma = 1.400$ for air.

(b) From Eq. (19.26), the work done is

$$W = \frac{1}{\gamma - 1}(p_1V_1 - p_2V_2)$$

Using $V_1/V_2 = 15.0$, we have

$$W = \frac{1}{1.400 - 1}\left[\begin{array}{l}(1.01 \times 10^5\ \text{Pa})(1.00 \times 10^{-3}\ \text{m}^3) \\ -\ (44.8 \times 10^5\ \text{Pa})\left(\dfrac{1.00 \times 10^{-3}\ \text{m}^3}{15.0}\right)\end{array}\right]$$

$$= -494\ \text{J}$$

We can check our result in part (b) by using Eq. (19.25). The number of moles of gas in the cylinder is

$$n = \frac{p_1V_1}{RT_1} = \frac{(1.01 \times 10^5\ \text{Pa})(1.00 \times 10^{-3}\ \text{m}^3)}{(8.314\ \text{J/mol} \cdot \text{K})(300\ \text{K})} = 0.0405\ \text{mol}$$

Then Eq. (19.25) gives

$$\begin{aligned} W &= nC_V(T_1 - T_2) \\ &= (0.0405\ \text{mol})(20.8\ \text{J/mol} \cdot \text{K})(300\ \text{K} - 886\ \text{K}) \\ &= -494\ \text{J} \end{aligned}$$

The work is negative because the gas is compressed.

Summary

- The First Law of Thermodynamics has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    1. **Thermodynamic systems**
    2. **Work done during volume changes**
    3. **Paths between thermodynamic states**
    4. **Internal energy and the first law of thermodynamics**
    5. **Kinds of thermodynamic processes**
    6. **Internal energy of an ideal gas**
- Textbook
  - Chapter 19: The First Law of Thermodynamics